

Subject : .....

No. : 19-0

Date : .....

|                    |                       |                                                                                    |
|--------------------|-----------------------|------------------------------------------------------------------------------------|
| 使估計 bias 的 problem | Causal<br>Confounding | Econometric<br>Endogeneity { Omitted var<br>Measurement Error<br>Simultaneous Bias |
| 可識別性和 正確推論假設       | Exchangeability       | Exogeneity { Weak<br>Strong                                                        |

No confounding given exchangeability of group compared

Def. The covariance of  $X$  and  $Y$  is

$$\text{Cov}(X, Y) = E[(X - E(X))(Y - E(Y))]$$

$$= E(XY) - E(X)E(Y)$$

Def. If  $X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$

not generally true

Ex.  $X \sim f(x-\theta)$  symmetric around 0  $\wedge E(X) = 0$

$Y = I_{\{|X-\theta| < 2\}}$

then

$$E(XY) = \int_{-\infty}^{\infty} x I_{\{|x-\theta| < 2\}} f(x-\theta) dx = \int_{-2}^2 (t+\theta) f(t) dt$$

$$= \theta \int_{-2}^2 f(t) dt = E(X)E(Y)$$

\* Property (Bilinear)

$$\begin{aligned} \text{Cov}(aX + bY, cW + dV) &= ac \text{Cov}(X, W) \\ &+ ad \text{Cov}(X, V) \\ &+ bc \text{Cov}(Y, W) \\ &+ bd \text{Cov}(Y, V) \end{aligned}$$

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A general model for the relationship between  $Y$  and  $X_1, \dots, X_m$

$$Y = f(X_1, \dots, X_m) + \varepsilon \quad (\text{规律} + \text{随机})$$

where  $f$  is unknown and arbitrary.

★ Consider the 1st order Taylor expansion

$$\begin{aligned} f(x) &\approx f(a) + \sum_{k=1}^m \frac{df}{dx_k}(a)(x_k - a_k) \\ &= \left[ f(a) - \sum_{k=1}^m \frac{df}{dx_k}(a)a_k \right] + \sum_{k=1}^m \left[ \frac{df}{dx_k}(a) \right] x_k \\ &\triangleq \beta_0 + \sum_{k=1}^m \beta_k x_k \end{aligned}$$

the local approx may be achievable by a LM.

★ In general,

$$Y = \underbrace{\sum_{k=0}^p g_k(X_1, \dots, X_m) \beta_k}_{\text{deterministic component}} + \underbrace{\varepsilon}_{\text{random component}} \quad \begin{cases} E(\varepsilon) = 0 \\ V(\varepsilon) = \sigma^2 \end{cases}$$

$g_k(X_1, \dots, X_m)$  is known fn of  $X_1, \dots, X_m$

Ex.  $X_1^k, X_1 X_2, \log\left(\frac{X_1}{X_2}\right)$

★ Matrix form

$$\underline{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \dots & x_{p1} \\ \vdots & \vdots & & \vdots \\ 1 & x_{1n} & \dots & x_{pn} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix} = \underline{X} \underline{\beta} + \underline{\varepsilon}$$

and  $E(\underline{\varepsilon}) = \underline{0}$

$$V(\underline{\varepsilon}) = \sigma^2 \underline{I}_n$$

★ LSE = define the best  $\hat{\underline{\beta}}$  as that minimize  $S(\underline{\beta})$

$$S(\underline{\beta}) = (\underline{y} - \underline{X} \underline{\beta})^T (\underline{y} - \underline{X} \underline{\beta})$$

$$\underset{\underline{\beta}}{\operatorname{argmin}} S(\underline{\beta}) \Leftrightarrow \frac{dS(\underline{\beta})}{d\underline{\beta}} = \underline{0}$$

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obtain Normal equation

suff stat of  $\beta$   
(under normal)

$$X^T X \beta = X^T y$$

assume  $X^T X$  is nonsingular

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

★ Property of  $\hat{\beta}$

i) Unbiasedness  $E(\hat{\beta}) = \beta + (X^T X)^{-1} E(X^T \epsilon) = \beta$

$\Downarrow$ , Exogeneity

ii) Variance irrelevant to  $y$  and  $\beta$ ,  $V(\hat{\beta}) = \sigma^2 (X^T X)^{-1}$

★ Orthogonal projection

$H = X(X^T X)^{-1} X^T$  is the orthogonal projection matrix onto the space spanned by  $X$

i) Idempotent  $H^2 = H$

ii) Symmetry  $H^T = H$

then

$$\hat{y} = X \hat{\beta} = H y$$

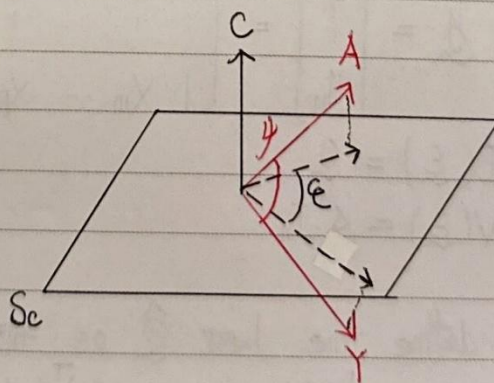
$$\hat{\epsilon} = y - \hat{y} = (I - H) y$$

$$\hat{\epsilon}^T \hat{\epsilon} = y^T (I - H) y = \epsilon^T (I - H) \epsilon$$

★ Partial Correlation (淨相關)

Consider  $A, Y, C$

$S_C$  is hyperplane  $\perp C$



$$\cos(\psi) = \text{Corr}(Y, A) = r_{YA}$$

$$\cos(\phi) = \text{Corr}(\hat{\epsilon}_Y, \hat{\epsilon}_A) = r_{YA|C}$$

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Formally, the partial correlation between  $A$  and  $Y$  given a set of  $m$  controlling variables  $\underline{C} = (C_1, \dots, C_m)$

is correlation between  $\hat{\epsilon}_Y$  and  $\hat{\epsilon}_A$  from

$$Y = \theta_0 + \theta_1 C_1 + \dots + \theta_m C_m + \epsilon_Y \quad \hat{\epsilon}_Y = Y - \hat{Y}$$

$$A = \theta_0 + \theta_1 C_1 + \dots + \theta_m C_m + \epsilon_A \quad \hat{\epsilon}_A = A - \hat{A}$$

Actually, the  $m$ th order partial correlation ( $|\underline{C}|=m$ ) can be easily computed by  $(m-1)$ th order partial correlations for any  $C^* \in \underline{C}$ , that

$$r_{Y|A|\underline{C}} = \frac{r_{Y|A|\underline{C} \setminus \{C^*\}} - r_{C^*|\underline{C} \setminus \{C^*\}} r_{C^*|A|\underline{C} \setminus \{C^*\}}}{(1 - r_{C^*|\underline{C} \setminus \{C^*\}}^2)^{\frac{1}{2}} (1 - r_{C^*|A|\underline{C} \setminus \{C^*\}}^2)^{\frac{1}{2}}}$$

Note in case,  $\underline{C} = (C_1, C_2)$

$$r_{Y|A|C_1, C_2} = \frac{r_{Y|A|C_1} - r_{C_2|C_1} r_{C_2|A|C_1}}{(1 - r_{C_2|C_1}^2)^{\frac{1}{2}} (1 - r_{C_2|A|C_1}^2)^{\frac{1}{2}}}$$

$$= \frac{r_{Y|A|C_2} - r_{C_1|C_2} r_{C_1|A|C_2}}{(1 - r_{C_1|C_2}^2)^{\frac{1}{2}} (1 - r_{C_1|A|C_2}^2)^{\frac{1}{2}}}$$

★ Regression and Partial Correlation

$$\begin{bmatrix} Y \\ X_1 \\ \vdots \\ X_p \end{bmatrix} \sim N(\underline{\mu}, \underline{\Sigma})$$

$$\underline{\mu} = \begin{bmatrix} \mu_Y \\ \mu_1 \\ \vdots \\ \mu_p \end{bmatrix}$$

$$\underline{\Sigma} = \begin{bmatrix} \Delta_Y^2 & \Delta_{Y1} & \dots & \Delta_{Yp} \\ \Delta_{1Y} & \Delta_1^2 & \dots & \Delta_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ \Delta_{pY} & \Delta_{p1} & \dots & \Delta_p^2 \end{bmatrix}$$

where  $\Delta_{jk} = \text{Cov}(X_j, X_k)$   $\Delta_{kk} = \Delta_k^2 = \text{Var}(X_k)$   
 $\Delta_{rk} = \text{Cov}(Y, X_k)$   $\Delta_Y^2 = \text{Var}(Y)$   
 $E(X_k) = \mu_k$   
 $E(Y) = \mu_Y$

the dist<sup>n</sup> of  $Y$  conditional on  $X = (X_1, \dots, X_p)$

$$Y|X \sim N(\mu^*, \Sigma^*)$$

the mean fn

$$\mu^* = E(Y|X) = \mu_y + \Gamma_{yx} \Gamma_{xx}^{-1} (X - \mu_x)$$

$$\triangleq \beta_0 + \sum_{k=1}^p \beta_k X_k$$

Express  $\beta_k$  in terms of partial correlation

Let  $\{a:b\} = \{a, a+1, \dots, b-1, b\}$   $a < b$

$\Omega(\{a:b\} \setminus \{c\})$  is arbitrary permutation of  $\{a:b\} \setminus \{c\}$

$\Omega(\{a:b\} \setminus \{c\})[p]$  is the first  $p$  elements of  $\Omega(\{a:b\} \setminus \{c\})$

$(\Omega(\{a:b\} \setminus \{c\})[p] = \emptyset \text{ when } p < 1)$

then

$$\beta_k = \frac{\Delta_Y}{\Delta_k} \left( \prod_{i \in \Omega(\{1:p\} \setminus \{k\})} \frac{dr_{i1} \Omega(\{1:p\} \setminus \{k\})[p(i)-1]}{dk_{i1} \Omega(\{1:p\} \setminus \{k\})[p(i)-1]} \right) r_{Yk | \Omega(\{1:p\} \setminus \{k\})}$$

where  $dr_{i1} = (1 - r_{Y1i}^2)^{\frac{1}{2}}$

position of element  $i$

Ex.  $p=2$

for  $k=1$ ,  $\Omega(\{1:2\} \setminus \{1\}) = \{2\}$

$$\beta_1 = \frac{\Delta_Y}{\Delta_1} \frac{dr_{12}}{d_{12}} r_{Y1|2}$$

$p=3$

for  $k=1$ ,  $\Omega(\{1:3\} \setminus \{1\}) = \{2,3\}$  or  $\{3,2\}$

(i) case  $\{2,3\}$

$$\beta_1 = \frac{\Delta_Y}{\Delta_1} \left( \frac{dr_{12} dr_{13}}{d_{12} d_{13|2}} \right) r_{Y1|23}$$

(ii) case  $\{3,2\}$

$$\beta_1 = \frac{\Delta_Y}{\Delta_1} \left( \frac{dr_{13} dr_{12|3}}{d_{13} d_{12|3}} \right) r_{Y1|32}$$

$p=1$

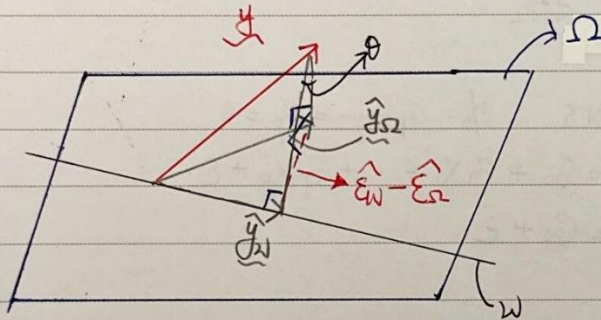
$$\beta_1 = \frac{\Delta_Y}{\Delta_1} \beta_1 = \frac{\text{Cov}(Y, X_1)}{\text{Var}(X_1)}$$

## \* Hypothesis Testing in Regression

The usual assumption  $\underline{\varepsilon} \sim N(\underline{0}, \sigma^2 \mathbf{I})$

Recall that

A geometric view of  $H_0 = \mathcal{W}$  vs  $H_1 = \Omega \setminus \mathcal{W}$


$$\hat{\epsilon}_w - \hat{\epsilon}_\Omega = (H_\Omega - H_w) \frac{y}{z}$$

the length

and  $RSS_W - RSS_2 \perp RSS_2$

- If  $RSS_H - RSS_\Omega$  is small,  $\omega$  is a more adequate model relative to  $\Omega$ .
- Suggest  $\frac{RSS_H - RSS_\Omega}{RSS_\Omega} (= \tan^2(\theta))$

the denominator is used for scaling.

Why not divided by  $RSS_n$ ?

$\Rightarrow A: 1) \text{ orthogonality}$

2) central chi-square under  $H_0$

Under  $H_0$ .

$$\frac{RSS_1 - RSS_2}{\sigma^2} \sim \chi^2_{(df_1 - df_2)}$$

$$\frac{RSS_2}{\sigma^2} \sim \chi^2_{(df_2)}$$

the general form

$$F = \frac{\frac{RSS_1 - RSS_2}{df_1 - df_2}}{\frac{RSS_2}{df_2}} \sim F_{df_1 - df_2, df_2}$$

a) Test of all predictors,  $H_0: \beta_1 = \dots = \beta_p = 0$ 

$$\Omega: Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \epsilon$$

$$\omega: Y = \beta_0 + \epsilon$$

b) Test one predictor,  $H_0: \beta_k = 0$ 

$$\Omega: Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \dots + \beta_p X_p + \epsilon$$

$$\omega: Y = \beta_0 + \beta_1 X_1 + \dots + \cancel{\beta_k X_k} + \dots + \beta_p X_p + \epsilon$$

c) Test pair of predictors,  $H_0: \beta_i = \beta_j = 0$ 

$$\Omega: Y = \beta_0 + \beta_1 X_1 + \dots + \beta_i X_i + \dots + \beta_j X_j + \dots + \beta_p X_p + \epsilon$$

$$\omega: Y = \beta_0 + \beta_1 X_1 + \dots + \cancel{\beta_i X_i} + \dots + \cancel{\beta_j X_j} + \dots + \beta_p X_p + \epsilon$$

d) Test a subspace/subset,  $H_0: \beta_j + \beta_k = 1$  or  $\beta_j = \beta_k$ 

$$\Omega: Y = \beta_0 + \beta_1 X_1 + \dots + \beta_j X_j + \dots + \beta_k X_k + \dots + \beta_p X_p + \epsilon$$

$$\omega: Y = \beta_0 + \beta_1 X_1 + \dots + \beta_j (X_j + X_k) + \dots + \cancel{\beta_k X_k} + \dots + \beta_p X_p + \epsilon$$

e) Linear Hypothesis Test,  $H_0: R\beta = c$ 

$$F = \frac{(R\beta - c)^T (R(X^T X)^{-1} R)^{-1} (R\beta - c) / g}{RSS / (n - k - 1)} \sim F_{g, n - k - 1}$$

rank = g

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What is Endogeneity?

When an explanatory variable is contemporaneously correlated w/ regression error one is said to have an endogeneity problem

$$\hat{Y} = X\hat{\beta} + \hat{\varepsilon}$$

$$\Downarrow$$

$$E(X\hat{\varepsilon}) \neq 0$$

Ex. In Simple Linear Regression

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

$$\begin{aligned} \text{Cov}(Y, X) &= \text{Cov}(\beta_0 + \beta_1 X + \varepsilon, X) \\ &= \beta_1 V(X) + \text{Cov}(\varepsilon, X) \end{aligned}$$

$$\Rightarrow \beta_1 = \frac{\text{Cov}(Y, X)}{V(X)} - \frac{\text{Cov}(\varepsilon, X)}{V(X)}$$

$$\Rightarrow \beta_1 = \frac{\text{Cov}(Y, X)}{V(X)} \quad \text{if} \quad \text{Cov}(\varepsilon, X) = 0$$

$$\Downarrow$$

$$E(\varepsilon X) - E(\varepsilon)E(X)$$

$$\Downarrow$$

$$E(X\varepsilon) = 0$$

★ Method of moment

The Law of Large Number

$$\frac{X_1 + \dots + X_n}{n} \xrightarrow{P} E(X)$$

Consider  $Y = g(X) = X^k$ 

$$\frac{Y_1 + \dots + Y_n}{n} \xrightarrow{P} E(Y) = E[g(X)] = E(X^k)$$

$$m_1 = \frac{\sum X_i}{n} \triangleq E(X) = g_1(\theta_1, \dots, \theta_k)$$

$$m_2 = \frac{\sum X_i^2}{n} \triangleq E(X^2) = g_2(\theta_1, \dots, \theta_k)$$

$$m_k = \frac{\sum X_i^k}{n} \triangleq E(X^k) = g_k(\theta_1, \dots, \theta_k)$$

Under assumption  $\begin{cases} E(\varepsilon) = 0 \\ E(X\varepsilon) = 0 \end{cases}$

by MME, we have

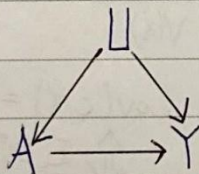
$$\begin{cases} \frac{\sum \varepsilon_i}{n} = \frac{\sum (y_i - \beta_0 - \beta_1 x_i)}{n} = 0 \\ \frac{\sum \varepsilon_i x_i}{n} = \frac{\sum (y_i - \beta_0 - \beta_1 x_i) x_i}{n} = 0 \end{cases}$$

$\Downarrow$

$$\begin{cases} \sum y_i = n\beta_0 + \beta_1 \sum x_i \\ \sum x_i y_i = \beta_0 \sum x_i + \beta_1 \sum x_i^2 \end{cases} \quad (\text{normal equation})$$

3 Common Source of Endogeneity

★ Omitted Variable



- If a confounder  $U$  is omitted in linear regression then  $A$  would become endogenous.
- Unexplained variable  $U$  leaks into the error term then  $A$  will be correlated w/ the error term.

Ex.

$$Y = \beta_0 + \beta_1 A + \beta_2 U + \varepsilon \quad (\text{True})$$

$$Y = \beta_0 + \beta_1 A + \tau \quad (\text{Est Model})$$

$$\tau = \beta_2 U + \varepsilon$$

$$\Rightarrow \text{Cov}(A, \varepsilon) = \beta_2 \text{Cov}(A, U) \neq 0$$

- Sign of  $\beta_2$  and Sign of  $\text{Cov}(A, U)$   
 $\Rightarrow$  Direction of the bias.